



BUYER'S GUIDE

WHAT IS A **STAIRCASE TOWER**?



A Scaffolding Staircase Tower is a heavy-duty, modular vertical access system designed to provide safe, temporary passage for workers and materials between different structural levels. Utilizing high-strength medium-class steel pipes and advanced connection nodes, it ensures secure access on intensive job sites.

Engineered with an premium anti-skid dimple platform finish, full double-sided guardrails, and secure anchorage tie bars, our towers are optimized for superior load-bearing capacity and fast, tool-free assembly. They completely replace unsafe vertical ladders across construction, industrial, and offshore projects.

DOWNLOAD DATASHEET PDF

Anti-Skid Dimple

Secure grip in wet & dusty conditions

380 kg UDL

Heavy-duty design platform capacity

Full Fall Protection

Guardrails & 150mm toe guards

Dual Finish Protection

Red-Oxide and Blue Enamel paint

WHY CHOOSE OUR STAIRCASE TOWERS?

Prime Steel Tech Staircase Towers provide industry-leading structural stability and worker safety. Built with a specialized Cuplock node connection system, a single cup locks up to 4 ledgers simultaneously for incredibly fast, tool-free assembly.

Every tower includes custom-engineered step arms with dimple-pressed anti-skid surfaces to prevent slips on-site. Equipped with double-sided handrails (top rails at 1,100 mm and mid-rails at 700 mm) alongside 150 mm high landing toe boards, these structures provide absolute edge protection. Backed by third-party ISO 9001 certifications, they offer premium, compliant safety standards that standard ladder systems simply cannot match.

PRIME STEEL TECH TOWER SPECIFICATIONS

PARAMETER	GENERAL SPECIFICATION
Primary Material	M.S. (IS 1161 Grade YST210) / Medium Class Pipe
Tube Outer Diameter	48.3 mm
Platform Height / Overall Tower Height	6,000 mm / 7,500 mm (incl. top rail)
Stair Width & Material	≥ 1,000 mm clear / M.S. Dimple Plate
Platform Design Load (UDL)	380 kg (3.73 kN/m² 75.7 psf)
Surface Protection Finish	Red-Oxide + Blue Enamel Paint
Components & Locking Type	Drop-Forged Steel Top Cup / Pressed Metal Bottom Cup

PARAMETER

GENERAL SPECIFICATION

Regulatory Compliance

IS 1161, IS 3696, IS 4014, EN 12811-1, BS 1139-3, OSHA

AVAILABLE TOWER CONFIGURATIONS

CONFIGURATION
PARAMETER

PSST-4 SYSTEM

PSST-8 SYSTEM

Number of Vertical
Legs

4 Legs

8 Legs

Grid Layout Size (L x
W)

2,500 x 2,000 mm

2,000 x 3,800 mm

Staircase Step Rise

214 mm (Class-B EN
12811-1)

187.5 mm (Class-B EN
12811-1)

Staircase Pitch/Angle

49° Angle

50° Angle

Landing Platform
Depth

600 mm (in direction of
travel)

800 mm (in direction of
travel)

Landing Platform
Width

2,000 mm

2,000 mm

FACTORS TO CONSIDER BEFORE BUYING STAIRCASE TOWERS

01. GRID SIZE & LAYOUT CONSTRAINTS

Assess the space limitations of your job site. The 4-leg configuration (PSST-4) utilizes a 2,500 x 2,000 mm footprint which is ideal for tight structural spacing, whereas the heavy-duty 8-leg configuration (PSST-8) requires a wider 2,000 x 3,800 mm footprint.

02. HEIGHT & ANCHORAGE REQUIREMENTS

Ensure your configuration aligns with standard heights or requests customized extensions. Remember that structural tie bars are mandatory once the total tower height exceeds 3 times the least base dimension, with specific wall attachment rules up to 4 m vertically.

03. UNIFORMLY DISTRIBUTED LOAD (UDL)

Evaluate the density of worker traffic and tool transportation. Prime Steel Tech towers feature a high platform UDL of 380 kg (3.73 kN/m²), strictly meeting Class-B global standard compliance for multi-worker movement.

04. COMPLIANCE & ON-SITE LEGISLATION

Verify that your safety code mandates align with local regulations. These towers conform strictly to regional Indian Standards (IS 3696/4014) mandated under the BOCW Act 1996, alongside global standardizations including EN 12811-1 and US OSHA.

INDUSTRIES USING STAIRCASE TOWERS

Civil Infrastructure

High-rise residential builds, bridges, flyovers, and metro viaduct networks.

Heavy Industrial Plants

Refineries, petrochemical processing zones, cement factories, and power hubs.

Marine Fabrication

Ship construction, shipyard maintenance, dockyard works, and offshore platforms.

Specialized Sectors

Nuclear energy sites, defence structures, and aircraft hangar staging.

PRIME STEELTECH

INDIA PVT LTD

Manufacturing high-quality Ringlock scaffolding, Cuplock scaffolding, anti-skid walkway planks, and safe staircase towers for India’s construction industry

SOLUTIONS

- Product Catalog
- Ringlock Scaffolding
- Staircase Towers
- Walkway Planks
- Staircase Tower Rentals NEW



COMPANY

RESOURCES

About Us

Live Quote Engine →

Our Projects NEW

Buyer's Guide NEW

Find Us / Locations NEW

Engineering Blogs

Contact Sales

Customer FAQs

© 2026 Prime Steeltech India PVT LTD. All rights reserved.

Developed By Dhruv and Ayush Mumbai, Maharashtra, India